

Messaging

As a comprehensive model of behavior change, the Transtheoretical Model of Behavior Change (TTM) has been widely applied to understanding and promoting individual health behaviors, such as examining college students' participation in physical activity (Han et al., 2017; Liu et al., 2018). The messages developed for this campaign directly leverage the Transtheoretical Model (TTM) to target the specific cognitive and behavioral processes relevant to each stage of behavior change (Prochaska & DiClemente, 1982; Velicer et al., 1999). TTM identifies five key stages: precontemplation, contemplation, preparation, action, and maintenance.

Precontemplation Stage's Message: Feeling Tired Again? Your Sleep May Be to Blame.

In this stage, students are generally unaware or dismissive of their problematic sleep behaviors and thus lack motivation to change. Therefore, the message needs to be used as the fulcrum for prying behavioral change. The primary goal is to gently increase students' awareness of the negative effects of bedtime procrastination. The message is non-threatening yet thought-provoking, initiating a cognitive shift that may prompt them to consider a change in the future.

Contemplation Stage's Message: Give Yourself the Gift of Good Sleep

In this stage, individuals are beginning to acknowledge the negative consequences of their current behavior and are weighing the pros and cons of making a change. These messages emphasize benefits that resonate strongly with students, nudging them toward recognizing the greater advantages of improved sleep compared to the transient pleasures of nighttime distractions.

Preparation Stage's Message: Switch Off for Better Sleep (OpenAI, 2025).

Students in the preparation stage are planning to make a change soon and require clear, practical guidance alongside encouragement to bolster their confidence. These actionable messages directly address students' need for clear, manageable strategies. By offering simple, concrete behaviors (e.g., turning off devices before bed), the campaign boosts students' self-efficacy, essential for moving forward into action.

Action Stage's Message: Rest Well, Stress Less.

Students have begun making tangible behavior changes but require ongoing reinforcement and affirmation of immediate positive outcomes to maintain motivation. Reinforcement of immediate and tangible benefits (reduced stress, increased daytime energy) helps solidify new behaviors. This positive feedback sustains motivation, encouraging students to continue their healthier bedtime routines.

Maintenance Stage's Message: Own Your Sleep, Unlock Your Best Self

In the maintenance stage, students have successfully changed their behavior for a sustained period but remain at risk for relapse. Messages should reinforce long-term benefits, provide strategies for maintaining gains, and encourage resilience against temptation. Messages in this stage highlight the enduring positive outcomes of maintained behavior, thereby reducing temptation and strengthening commitment. Reinforcing students' self-efficacy and celebrating their progress helps solidify long-term change, a critical component of TTM.

Communication Channels

The campaign will be implemented through three main channels to reach the undergraduate student population at Johns Hopkins University, utilizing an integrated approach of both online and offline strategies to maximize communication effectiveness.

First, the campaign will launch a Healthy Sleep Newsletter Program: *Goodnight Guide* (OpenAI, 2025) via email. Students who subscribe will receive weekly newsletters offering sleep health tips and reminders. Email interventions provide a direct and personal mode of communication and, when designed to be user-friendly and low-barrier, have proven highly effective in promoting health behavior change (van Genugten et al., 2016). Sending weekly updates through student subscription accounts ensures both broad coverage and targeted messaging. The consistent exposure provided by a weekly newsletter is critical for sustaining behavior change over time.

Second, the campaign will utilize the university's organization Instagram account @jhuhop (<https://www.instagram.com/jhuhop/>) and JHU well-being account @jhuwellbeing (<https://www.instagram.com/jhuwellbeing/>) to disseminate sleep health information. Visually appealing posters and short infographics will be designed to promote the Healthy Sleep Program and encourage student subscriptions. The endorsement through the official school account will enhance the credibility and trustworthiness of the information shared. Instagram is particularly effective for reaching young adults due to its visual-centered design and high daily usage among college students. Health communication research has shown that visually engaging content (such as infographics and posters) generates higher levels of attention and behavior change compared to purely text-based messages (Maher et al., 2022). By leveraging Instagram posts and stories, the campaign aims to increase program visibility, normalize positive sleep behaviors, and drive student engagement.

Finally, offline activities will serve as an important channel for promoting behavior change. The campaign team will organize monthly interactive workshops where students can learn about the consequences of bedtime procrastination (BP) and practical strategies to address it, with small gifts offered as participation incentives. Face-to-face interaction remains crucial for reinforcing health behaviors. For example, the BED-PRO intervention, which integrated motivational interviewing with TTM, demonstrated that offline sessions contributed to a 63.8% reduction in BP duration and improved sleep efficiency (Suh et al., 2022). Offline workshops foster active learning, personal interaction, and reinforcement of behavior change strategies, all of which are critical for deepening commitment to healthy habits (Hayes et al., 2022). These sessions create an environment for immediate feedback, peer support, and experiential learning, further enhancing the effectiveness of the campaign.

Creative





Figure 1&2. Precontemplation stage's Instagram post example

(Note. Blue Jay illustration generated by OpenAI.)

The First Instagram post targets JHU students in the precontemplation stage. According to TTM, individuals in this stage are unaware of their problematic behaviors. The post highlights the negative consequences of bedtime procrastination (e.g., fatigue, weakened focus, health risks) reflects the decisional balance concept in TTM, where individuals weigh the pros and cons of behavior change (Prochaska & Velicer, 1997). This post aims to use a relatable problem description to raise awareness and prompt self-recognition of sleep procrastination. Additionally, publishing through a widely used platform like Instagram increases user reach (Maher et al., 2022) and delivers an early, non-confrontational nudge toward recognizing the need for change.

In *Figure 2*, the Zzzential Sleep Workshop creates a low-pressure entry point for students to start thinking about their sleep behaviors. Through interactive learning about BP's negative impacts, the workshop supports early-stage consciousness raising -- a critical TTM process that helps move individuals toward contemplating behavior change. Moreover, offering free lunch also reduces participation barriers, applying TPB's Perceived Behavioral Control principle (Ajzen, 1991), which makes it easier for students to follow through.



Figure 3. Preparation stage's email poster example

The poster would be attached to the *Goodnight Guide* email newsletter, which targets students who are beginning to recognize the importance of sleep and are preparing for behavior change. Aligned with TTM's focus on small, actionable steps at this stage (Prochaska & Velicer, 1997), the poster presents three practical tips. Each tip lowers the barriers to action by offering simple and realistic behaviors, supporting TPB's emphasis on increasing perceived behavioral control (Ajzen, 1991). By making the initial steps achievable and positively framed, the poster aims to build self-efficacy and intention, encouraging students to move confidently toward better sleep routines.

Additionally, distributing the content via the university's *Goodnight Guide* newsletter extends reach and ensures repeated exposure, which a key mechanism for strengthening behavior change (van Genugten et al., 2016).

Zzzential Sleep Workshop

Good sleep isn't just about feeling rested -- it's about unlocking your full potential.

Register now!

- What keeps you stuck in late-night habits, and how to break them
- Why better sleep boosts your focus, mood, and health
- Real strategies you can start tonight for lasting change



3:00~4:00 P.M. Wed
07/02/2025 Homewoods

Pick and Join!

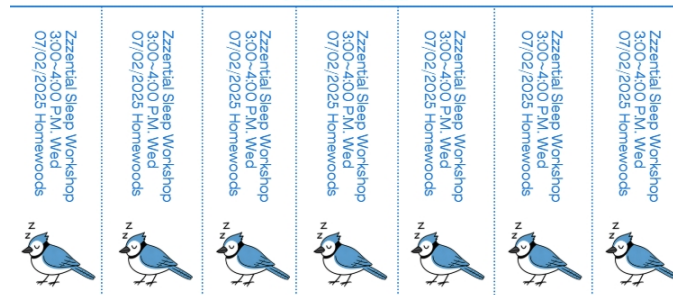


Figure 4. Contemplation stage's flyer example

(Note. Blue Jay illustration generated by OpenAI.)

The flyer targets students who are beginning to acknowledge that bedtime procrastination may be affecting their well-being but have not yet committed to change. According to TTM, individuals in the contemplation stage benefit from understanding both the risks of inaction and the benefits of change (Prochaska & Velicer, 1997). The flyer uses a motivational but low-pressure tone, aiming to move students from thinking about change toward actively preparing for it. These elements are designed to support decisional balance, helping students weigh the benefits of change more heavily than the perceived barriers. Additionally, the tear-off strips at the bottom of the flyer provide an easy, physical reminder of the workshop details. This feature will encourage commitment to attend.